Personal information

Websites & social links

- Dr. Artiom Kovnatsky | Portfolio & Blog
- Google Scholar
- LinkedIn
- Systemic Verification Engineering (SVE)
- S.V.E. [zenodo]

Keywords

Systemic Verification, AI Governance, Hybrid AI Architectures, Explainable AI (XAI), AI Systems Architecture, Trust & Risk Modeling, Probabilistic Systems, Knowledge Graphs, Decision Systems, Risk Engineering, Systems Thinking, Applied Mathematics, Differential Geometry, Computational Philosophy

Countries

Germany

Biography

https://www.artiomkovnatsky.com/#about

Activities

Collapse all

Works (24) Sort

S.V.E. 0 (1): The Epistemological Boxing Protocol

2025-12-31 | Preprint
DOI: 10.5281/ZENODO.18108307
CONTRIBUTORS: Artiom Kovnatsky

Show less detail

Language
English

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https://zenodo.org/doi/10.5281/zenodo.18108307

Contributors
Artiom Kovnatsky (Author)

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Show citation

Description
This paper introduces the Epistemological Boxing Protocol (EBP), a core methodological component of the Systemic Verification Engineering (SVE) framework.

EBP is a structured, AI-assisted adversarial protocol designed for collaborative truth-seeking and cognitive training in high-stakes domains. It reframes argumentation from winner-oriented debate to iterative purification of a falsifiable thesis via structured challenge, virtuous concession, and quantitative integrity assessment.

The paper presents the philosophical foundations, protocol architecture, seven-round structure, and computational model based on vectorial purification. This work is an architectural and methodological skeleton intended to enable reproducible implementation, critique, and extension.

Status: work-in-progress / preprint. Feedback and adversarial testing are explicitly encouraged.

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Source: Artiom Kovnatsky

S.V.E. 0 (2): The Socratic Investigative Process (SIP)

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DOI: 10.5281/ZENODO.18108751
CONTRIBUTORS: Artiom Kovnatsky

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English

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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint introduces the Socratic Investigative Process (SIP), an iterative, multi-agent epistemic protocol for computational truth approximation in complex, high-stakes systems.

SIP formalizes Socratic dialogue as a reproducible purification process on a semantic manifold, producing versioned, auditable "Iterative Facts". The framework extends to Meta-SIP, enabling synthesis across multiple independent investigations.

This paper is part of the Systemic Verification Engineering (SVE) program and represents a methodological skeleton intended for public audit and further development.

Repository: https://github.com/skovnats/SVE-Systemic-Verification-EngineeringLicense: SVE Public License v1.3

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S.V.E. I: The Theorem of Systemic Failure

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English

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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint introduces the Disaster Prevention Theorem, a socio-probabilistic model diagnosing the structural causes of catastrophic errors in modern collective decision-making systems.

Using the "Wisdom of the Crowds" paradigm and the metaphor of "guessing the ox's weight," the paper demonstrates why information architectures based on centralized, mediated authority are structurally unstable when addressing complex, high-stakes problems. It proves that an Independent Verification Mechanism (IVM) is a necessary and sufficient condition for restoring collective intelligence.

The work presents the computational architecture of such an IVM, grounded in a two-stage vector purification protocol powered by the Socratic Investigative Process (SIP). Psychological foundations (groupthink countermeasures), an economic ROI framework for verification, and a systematic red-teaming analysis of the mechanism itself are included.

This paper serves as the foundational diagnostic layer of the Systemic Verification Engineering (SVE) framework.

Status: work in progress / preprint. This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. II: The Architecture of Verifiable Truth

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CONTRIBUTORS: Artiom Kovnatsky

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Homepage URL
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Artiom Kovnatsky (Author)

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Description
This preprint presents the core architectural layer of Systemic Verification Engineering (SVE), an applied framework designed to address the structural crisis of trust in modern institutions.

Building on the Disaster Prevention Theorem, the paper introduces a three-stage verification architecture that explicitly separates factual analysis from value-based judgment. The protocol combines AI-assisted factual boundary setting ("Caesar's realm"), plural expert interpretation, and informed collective decision-making ("God's realm").

At its computational core lies the Epistemological Boxing Protocol and the Socratic Investigative Process, enabling adversarial purification of narratives and reproducible approximation of truth under conditions of uncertainty.

A concrete instantiation, System-Purgatory, is presented as a transparent alternative to traditional peer review, addressing known failure modes such as opacity and groupthink.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

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S.V.E. III: The Protocol for Academic Integrity

2025-12-31 | Preprint
DOI: 10.5281/ZENODO.18108907
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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint presents SYSTEM-PURGATORY, a concrete instantiation of the Systemic Verification Engineering (SVE) framework designed to address the reproducibility crisis and structural trust failures in modern academia.

The protocol transforms traditional peer review into a transparent, adversarial yet constructive Socratic process between a human author and an AI antagonist, arbitrated by a tri-judge AI panel. The process produces publicly auditable artifacts, including a complete dialogue transcript, a synthetic verification report, and a quantitative Integrity Score derived from iterative vectorial purification of the research thesis.

The paper details the protocol architecture, its antifragile design principles, governance and incentive re-engineering, and an economic analysis demonstrating the high return on investment (ROI) of verifiable science. It reframes scientific practice from opaque gatekeeping toward reproducible, integrity-centered verification.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. IV: The Beacon Protocol

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CONTRIBUTORS: Artiom Kovnatsky

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Contributors
Artiom Kovnatsky (Author)

Citation (bibtex)
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Description
This preprint introduces the Beacon Protocol, an ethical navigation framework within the Systemic Verification Engineering (SVE) program, designed to address ethical deadlocks and situations of radical uncertainty where data-driven verification alone is insufficient.

The paper proposes a geometric model of reality as a composite manifold integrating semantic, psychological, and physical spaces. Within this framework, ethical decision-making is modeled as navigation along geodesic paths that optimize long-term societal well-being under uncertainty.

A central construct, the "Christ-vector," is introduced in the paper as an explicit ethical process model derived from Christian moral principles. The framework does not impose belief; it formalizes these principles as a falsifiable navigational hypothesis analogous to optimization principles in physics.

Applications include the deconstruction of classical ethical dilemmas (e.g., the Trolley Problem) and the concept of Geopolitical Maieutics for conflict resolution through radical transparency and verification.

This publication represents a methodological and conceptual skeleton intended for public audit, critique, and long-term empirical evaluation.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. IX: Systemic Verification Engineering

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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint presents S.V.E. IX, an integrated synthesis of the Systemic Verification Engineering (SVE) framework, unifying epistemic verification, ethical navigation, and collective intelligence into a single coherent architecture.

The paper integrates three core components: (1) the Disaster Prevention Theorem, establishing the necessity of Independent Verification Mechanisms (IVM) for reliable collective decision-making; (2) the Beacon Protocol, modeling ethical navigation as geodesic optimization under uncertainty; and (3) a mathematical framework for semantic and cultural transformation using high-dimensional vector spaces.

The framework formalizes truth approximation, ethical decision-making, and institutional integrity as coupled dynamical processes subject to falsification, red-teaming, and measurable performance criteria. Applications include AI alignment, policy verification, conflict resolution, governance design, and protection against information warfare.

This publication represents an integrative methodological and conceptual skeleton intended for public audit, critique, and iterative empirical refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. V: An Operating System for Verifiable Democracy

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CONTRIBUTORS: Artiom Kovnatsky

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Description
This preprint presents S.V.E. V, a practical blueprint translating the Systemic Verification Engineering (SVE) framework into an implementable operating system for democratic governance.

Building on the Disaster Prevention Theorem and the three-stage architecture separating facts from values, the paper proposes an "Operating System for Verifiable Democracy." Core components include a citizen-driven verification service ("Fakten-TÜV"), an AI-powered radical transparency interface ("Socrates Bot"), and a decision-making protocol designed to maximize collective intelligence under conditions of uncertainty.

The system is analyzed through antifragile design principles and red-teaming of potential failure modes, including institutional capture, weaponized uncertainty, and adversarial suppression. An economic model ("ROI of Truth") demonstrates that verification infrastructure can yield orders-of-magnitude returns by preventing catastrophic policy errors.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. VI: The Protocol for Cognitive Sovereignty

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English

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Description
This preprint introduces the Protocol for Cognitive Sovereignty, an applied extension of the Systemic Verification Engineering (SVE) framework addressing the central strategic challenge of the 21st century: the degradation of collective sense-making under conditions of information warfare and systemic groupthink.

The paper diagnoses the structural vulnerabilities of modern governance—wicked problems, cognitive manipulation, and elite groupthink—and presents the Epistemological Boxing Protocol as an engineered institutional antidote. The protocol functions as a cognitive immune system, enabling adversarial stress-testing of national strategies, inter-agency conflict resolution, and the formation of antifragile, evidence-grounded policy.

A concrete implementation blueprint is proposed through the creation of a Center for Strategic Verification, detailing mandate, organizational structure, phased deployment, and an economic analysis demonstrating an exceptional return on investment (ROI) through the prevention of catastrophic strategic failures.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. VII: Hybrid Models of State Structure

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Description
This preprint presents S.V.E. VII, an applied framework for designing antifragile state structures that transcend the classical dichotomy between top-down hierarchy and bottom-up self-organization.

Within the Systemic Verification Engineering (SVE) paradigm, the paper introduces four hybrid governance models—the Consultative Network, Project Authoritarianism, Federation of Guilds, and State-Organism—each combining hierarchical stability with decentralized adaptability. The Epistemological Boxing Protocol functions as the core mechanism for feedback, conflict resolution, and truth-seeking across all models.

The framework incorporates systemic insurance mechanisms such as Skin in the Game, the Right to Neat Chaos, protected dissent ("Holy Fools"), and parallel structures, ensuring resilience against groupthink, capture, and strategic failure.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. VIII: Provisional Ontology for Verifiable Ethics and Knowledge

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Artiom Kovnatsky (Author)

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Description
This preprint introduces S.V.E. VIII, a provisional ontological hypothesis underlying the Systemic Verification Engineering (SVE) program.

The paper proposes that ethical and epistemic dynamics can be treated as a geometry, forming a working ontological assumption rather than an asserted truth. This ontology is explicitly exposed, falsifiable, and contingent: it exists only insofar as the models instantiated within it survive empirical and adversarial verification.

S.V.E. VIII formalizes a recursive loop in which reality informs an ontological hypothesis, which generates language and primitives, enabling models that are subjected to verification protocols. Failure of instantiated models is treated as evidence against the ontology itself, triggering revision of both the ontological layer and its linguistic representation.

The framework is presented as a meta-level control system governing the evolution of SVE protocols (SVE 0–VII), ensuring that neither language nor ontology becomes dogmatic or immune to falsification.

This publication represents a conceptual and methodological skeleton intended for public audit, critique, and long-term empirical evaluation.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. X: Cognitive Operating Systems for LLMs

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Contributors
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Description
This preprint introduces the Cognitive Operating System (CogOS) paradigm for Large Language Models (LLMs), reframing LLMs as general-purpose cognitive hardware requiring structured operating systems to achieve reliable, verifiable, task-specific cognition.

The paper presents the Triple Architect framework as a concrete CogOS implementation, integrating three complementary personas: Socrates (formal logic and falsification), Solomon (ethical arbitration and value assessment), and Ivan the Fool (humility and empathetic delivery). The architecture operates through five core mechanisms, including humility calibration with Dunning–Kruger correction, Bayesian prior elicitation, a five-column verification table separating facts, models, and values, dual Socratic feedback loops for mutual correction, and four-dimensional growth tracking.

The framework demonstrates applicability across strategic analysis, intellectual self-audit, education, and collaborative knowledge creation. Integrated within the broader Systemic Verification Engineering (SVE) program, S.V.E. X provides the operational layer translating abstract verification principles into executable cognitive processes for human–AI collaboration.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. XI: The Ox's Weights

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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint introduces S.V.E. XI, a concrete architecture for collaborative truth approximation based on Verifiable Knowledge Bases (VKBs) and Distributed Independent Verification Mechanisms (IVMs).

Building on the Disaster Prevention Theorem, the paper revisits Francis Galton's "ox weighing" experiment and formalizes the conditions under which collective intelligence can reliably converge toward truth. The proposed VKB architecture represents knowledge as a directed acyclic graph (DAG) of verified Socratic Investigative Processes (SIPs) and Meta-SIPs, each passing a three-stage verification pipeline: AI-assisted structured reasoning (SIP), adversarial stress-testing via Epistemological Boxing (EBP), and multi-reviewer human peer evaluation.

The system integrates DAO-based governance for foundational context databases (Patterns of Thinking and Values/Anti-Values), enabling transparency, resistance to capture, and dynamic refinement. Detailed implementations are provided for large-scale applications including Stack Overflow 2.0, Wikipedia Reformation, decentralized fact-checking infrastructure, and expert knowledge marketplaces.

This publication represents a methodological and architectural skeleton intended for public audit, critique, and iterative refinement.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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S.V.E. XII: The SYSTEM

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English

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Contributors
Artiom Kovnatsky (Author)

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Description
This preprint presents S.V.E. XII, the integrative capstone of the Systemic Verification Engineering (SVE) series. The paper consolidates the full SVE architecture into a unified system combining epistemic verification, ethical constraint navigation, and collective intelligence into a coherent, reproducible framework.

S.V.E. XII formalizes the interaction between Independent Verification Mechanisms (IVMs), Verifiable Knowledge Bases (VKBs), and cognitive operating systems for human–AI collaboration. It introduces a system-level view in which truth approximation, ethical decision-making, and institutional reliability are modeled as coupled processes subject to adversarial testing, transparency constraints, and iterative refinement.

The framework is designed for high-stakes domains, including AI alignment, governance, policy analysis, conflict resolution, and large-scale information systems. Rather than proposing new truths, the system specifies verifiable procedures for approaching truth under uncertainty and disagreement.

This document serves as an architectural and methodological skeleton intended for public audit, critique, and empirical extension.

Status: work in progress / preprint.

License: S.V.E. Public License v1.3 Repository: https://github.com/skovnats/SVE-Systemic-Verification-Engineering

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Artiom Kovnatsky (Author)

Citation (bibtex)
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Description
Systemic Verification Engineering (S.V.E.) v0.3 Release.

Reproducible epistemic protocols for high-stakes systems.

"Git for Thinking": building reproducible and transparent hybrid-AI verification and trust infrastructure for complex systems in high-stakes domains — AI, governance, science and policy.
Already applied in real-world civic and media stress tests (2025–).

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MADMM: A Generic Algorithm for Non-smooth Optimization on Manifolds

Lecture Notes in Computer Science
2016 | Book chapter
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Part of ISBN: 9783319464534
Part of ISBN: 9783319464541
Part of ISSN: 0302-9743
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CONTRIBUTORS: Artiom Kovnatsky; Klaus Glashoff; Michael M. Bronstein

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Multimodal Manifold Analysis by Simultaneous Diagonalization of Laplacians

IEEE Transactions on Pattern Analysis and Machine Intelligence
2015-12-01 | Journal article
DOI: 10.1109/tpami.2015.2408348
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CONTRIBUTORS: Davide Eynard; Artiom Kovnatsky; Michael M. Bronstein; Klaus Glashoff; Alexander M. Bronstein

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Gamut mapping with image Laplacian commutators

2014 IEEE International Conference on Image Processing (ICIP)
2014-10 | Journal article
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Laplacian colormaps: a framework for structure-preserving color transformations

Computer Graphics Forum
2014-05 | Journal article
DOI: 10.1111/cgf.12295
Part of ISSN: 0167-7055
Part of ISSN: 1467-8659
CONTRIBUTORS: D. Eynard; A. Kovnatsky; M. M. Bronstein

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Language
English

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Description
<jats:title>Abstract</jats:title><jats:p>Mappings between color spaces are ubiquitous in image processing problems such as gamut mapping, decolorization, and image optimization for color-blind people. Simple color transformations often result in information loss and ambiguities, and one wishes to find an image-specific transformation that would preserve as much as possible the structure of the original image in the target color space. In this paper, we propose Laplacian colormaps, a generic framework for structure-preserving color transformations between images. We use the image Laplacian to capture the structural information, and show that if the color transformation between two images preserves the structure, the respective Laplacians have similar eigenvectors, or in other words, are approximately jointly diagonalizable. Employing the relation between joint diagonalizability and commutativity of matrices, we use Laplacians commutativity as a criterion of color mapping quality and minimize it w.r.t. the parameters of a color transformation to achieve optimal structure preservation. We show numerous applications of our approach, including color-to-gray conversion, gamut mapping, multispectral image fusion, and image optimization for color deficient viewers.</jats:p>

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2025-12-31

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2025-12-31

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Coupled quasi-harmonic bases

Computer Graphics Forum
2013-05 | Journal article
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CONTRIBUTORS: A. Kovnatsky; M. M. Bronstein; A. M. Bronstein; K. Glashoff; R. Kimmel

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Description
<jats:title>Abstract</jats:title><jats:p>The use of Laplacian eigenbases has been shown to be fruitful in many computer graphics applications. Today, state-of-the-art approaches to shape analysis, synthesis, and correspondence rely on these natural harmonic bases that allow using classical tools from harmonic analysis on manifolds. However, many applications involving multiple shapes are obstacled by the fact that Laplacian eigenbases computed independently on different shapes are often incompatible with each other. In this paper, we propose the construction of common approximate eigenbases for multiple shapes using approximate joint diagonalization algorithms, taking as input a set of corresponding functions (e.g. indicator functions of stable regions) on the two shapes. We illustrate the benefits of the proposed approach on tasks from shape editing, pose transfer, correspondence, and similarity.</jats:p>

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Source: Artiom Kovnatsky

Geometric and Photometric Data Fusion in Non-Rigid Shape Analysis

Numerical Mathematics: Theory, Methods and Applications
2013-01-01 | Journal article
DOI: 10.4208/nmtma.2013.mssvm11
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Photometric Heat Kernel Signatures

Lecture Notes in Computer Science
2012 | Book chapter
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CONTRIBUTORS: Artiom Kovnatsky; Michael M. Bronstein; Alexander M. Bronstein; Ron Kimmel

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Stable Spectral Mesh Filtering

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